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(54) **PHARMACEUTICAL CRYSTAL OF CONTEZOLID ACEFOSAMIL, PREPARATION METHOD THEREFOR, AND USES THEREOF**

PHARMAZEUTISCHER KRISTALL VON CONTEZOLID ACEFOSAMIL,
HERSTELLUNGSVERFAHREN DAFÜR UND VERWENDUNGEN DAVON

PRODUIT PHARMACEUTIQUE CRISTALLIN DE CONTEZOLID ACÉFOSAMIL, SON PROCÉDÉ DE PRÉPARATION ET SES APPLICATIONS

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- **STEPHEN BYRN ET AL: "Pharmaceutical Solids: A strategic Approach to Regulatory Considerations", PHARMACEUTICAL RESEARCH, SPRINGER NEW YORK LLC, US, vol. 12, no. 7, 1 July 1995 (1995-07-01), pages 945-954, XP000996386, ISSN: 0724-8741, DOI: 10.1023/A:1016241927429**
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Description

TECHNICAL FIELD

[0001] The present invention relates to a pharmaceutical crystal of contezolid acefosamil, a preparation method thereof, and uses thereof.

BACKGROUND

[0002] Oxazolidinone compounds are a new class of synthetic compounds with inhibitory activity against various pathogenic microorganisms. Linezolid, the first drug of this class, has been approved for the treatment of Gram-positive infections. Linezolid has good antibacterial activity, but just as indicated in the "warnings" section of the prescription labeling of linezolid, myelosuppression and monoamine oxidase inhibition are major factors that limit the use of linezolid. Chinese Patent Publication No. CN105612166A discloses that Conzolid is a novel oxazolidinone compound with excellent safety features and good antibacterial activity.

[0003] In addition to the required antibacterial activity and safety, effective antibacterial drugs must have solubility and stability suitable for practical use. Chinese Patent Publication No. CN105612166A discloses a therapeutic water-soluble (O-carbonyl) phosphoramidate prodrug, which has good water solubility. However, the products obtained according to the preparation method of contezolid acefosamil provided by this publication are amorphous powders which are unstable and often lead to decomposition, wherein the X-ray powder diffraction analysis (FIG. 1) does not show any diffraction peak between $0\sim 40^\circ 2\theta$, no matter whether the products are obtained via laboratory's traditional operations such as extraction, evaporation under reduced pressure, or lyophilized by HPLC purification.

[0004] WO 2015/127316 A1 provides water-soluble O-carbonyl phosphoramidate prodrugs, or pharmaceutically acceptable salts thereof that are therapeutic or antibacterial agents. WO 2015/127316 A1 also provides pharmaceutical compositions containing the prodrugs, methods for their use, and reagents and methods for preparing these compounds.

[0005] Bryn et al. ("Pharmaceutical Solids : A Strategic Approach to Regulatory Considerations" Pharmaceutical Research, Vol. 12, No. 7, 1995, pages 945-954) describes a conceptual approach to the characterisation of pharmaceutical solids. In particular, the review provides four flow charts which are suggested as tools to develop information on pharmaceutical solids for both scientific and regulatory purposes.

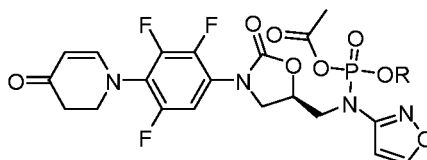
[0006] Caira (Montchamp Jean-Luc: "Crystalline Polymorphism of Organic Compounds", Topics in Current Chemistry, Vol. 198, 1 January 1998, pages 163-208) explains how crystal polymorphism is encountered in all areas of research involving solid substances. Caira presents a summary of thermodynamic, kinetic and structural considerations of polymorphism and surveys a wide variety of techniques appropriate to the study of organic crystalline polymorphism and pseudopolymorphism.

[0007] Hilfiker ("Polymorphism in the Pharmaceutical Industry", ISBN: 3-527-31146-7, Pages 1-19) discusses the relevance of solid-state properties for pharmaceutical products.

SUMMARY

[0008] In view of the above-mentioned problems, the present invention provides a crystal or crystal complex of contezolid acefosamil and a preparation method thereof, which has a significantly improved stability as compared to the amorphous powder, and thus can meet the requirements of pharmaceutical agents in clinical use.

[0009] In one aspect, the present invention relates to a crystal or crystal complex of formula I, wherein R is sodium. ONa and O^-Na^+ are equivalent.



I

The crystal or crystal complex exhibits diffraction peaks at $6.9^\circ \pm 0.2^\circ$, $14.7^\circ \pm 0.2^\circ$, $15.5^\circ \pm 0.2^\circ$, $16.5^\circ \pm 0.2^\circ$, $20.2^\circ \pm 0.2^\circ$, $22.9^\circ \pm 0.2^\circ$, and $23.7^\circ \pm 0.2^\circ$ 2θ in its X-ray powder diffraction (XRPD) pattern.

[0010] The crystal or crystal complex provided by the present invention greatly improves the stability of contezolid acefosamil, which is beneficial to the production and transportation of the drug, and the routine delivery of the drug to

patients or mammals in need of treatment for bacterial infections, and thereby ensuring the clinical use of the new safe and effective oxazolidinone.

[0011] Preferably, wherein the crystal or crystal complex has a melting point of 220 ± 10 °C.

[0012] Preferably, the crystal or crystal complex contains less than 6% of sodium by weight.

[0013] In the second aspect, the present invention provides a method for preparing the above-mentioned crystal or crystal complex, comprising the following steps:

(1) preparing the compound of formula I as a crude product;

(2) dissolving the compound of formula I as a crude product in a mixed solution of an organic solvent and water;

(3) adding inorganic salt to the mixture, separating the product phase to remove part or all of the volatile substances, and adding a second organic solvent, so as to obtain a crystal via crystallization; and

(4) separating out the crystal of the compound of formula I, wherein in step (2), the organic solvent is selected from at least one of acetonitrile, acetone, methanol, ethanol, propanol, isopropanol, butanol, and tetrahydrofuran, wherein in step (3), the inorganic salt is selected from at least one of sodium chloride, sodium bromide, sodium iodide, sodium citrate, sodium dihydrogen phosphate, disodium hydrogen phosphate, sodium phosphate, sodium sulfate, sodium carbonate, sodium bicarbonate, sodium acetate and their hydrates, and wherein in step (3), the second organic solvent is selected from at least one of acetonitrile, acetone, ethanol, and isopropanol.

[0014] Preferably, in step (2) the volume ratio of the organic solvent and water is 1:9 to 4:6.

[0015] The disclosure provides a use of the above-mentioned crystal or crystal complex for preparing an antibiotic medicine.

[0016] The disclosure provides a method for treating infections with above-mentioned crystal or crystal complex.

[0017] The disclosure provides a use of the above-mentioned preparation methods for preparing antibiotic agents.

[0018] In a third aspect, the present invention provides a pharmaceutical composition comprising the above-mentioned crystal or crystal complex and a pharmaceutically acceptable carrier.

[0019] In a fourth aspect, the present invention provides a crystal or crystal complex according to the first aspect or a pharmaceutical composition according to the third aspect for use in the treatment of a microbial infection in a mammal.

[0020] Preferably, the crystal, crystal complex or pharmaceutical composition is for use in treating a bacterial infection in a human.

[0021] Preferably, the bacterial infection is a Gram-positive bacterial infection.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022]

FIG. 1 shows an X-ray powder diffraction analysis of Product form 2 in Example 1.

FIG. 2 shows an X-ray powder diffraction analysis of Product 3 form (crystal) in Example 1.

FIG. 3 shows an X-ray powder diffraction analysis of product 1 form in Example 1.

FIG. 4 shows a differential thermal scanning analysis of Product 3 form (crystal) in Example 1.

FIG. 5 shows a differential thermal scanning analysis of the product obtained by re-dissolving the crystal in water and then being lyophilized.

DETAILED DESCRIPTION

[0023] Hereinafter, the present invention will be further described with the following examples. It should be understood that the following examples are used to explain this invention and do not mean to limit the scope of this invention.

The terms

[0024] The term "about" when used in relation to the peak position of the X-ray powder diffraction pattern refers to the inherent variability of the peak, which depends on, for example, the calibration of the used equipment, the method for producing the poly crystal, the life of crystalline material, etc., depends on the instrument used. In this case, the measurement variability of the instrument is about $\pm 0.2^\circ 2\theta$. Those skilled in the art can understand the use of "about" in this context.

[0025] The term "mammal" refers to all mammals including humans, livestock, and pets.

[0026] "Treating a disease" or "treatment of a disease" includes: (1) preventing the disease, *i.e.*, causing the clinical symptoms of the disease not to develop in a mammal that may be exposed to or predisposed to the disease but does not yet experience or display symptoms of the disease, (2) inhibiting the disease, *i.e.*, arresting or reducing the devel-

opment of the disease or its clinical symptoms, or (3) relieving the disease, *i.e.*, causing regression of the disease or its clinical symptoms.

[0027] A "therapeutically effective amount" means the amount of a compound that, when administered to a mammal for treating a disease, is sufficient to effect such treatment for the disease. The therapeutically effective amount will vary depending on the compound, the disease and its severity and the age, weight, etc., of the mammal to be treated.

[0028] The term "crystal complex" refers to a pharmaceutically acceptable crystal mixture containing a certain crystal. The crystal content can account for 50% to 90% of a product also including amorphous materials, moisture, and solvated organic solvents.

[0029] The term "complex" refers to a mixture or co-crystal.

[0030] The term "solvate" of a compound means that the compound contains a certain pharmaceutically acceptable solvent, whether chemically bound, such as a crystalline solvate, or remains, or as an additive, such as in a wet or lubricious solid.

[0031] The term "dissolving" refers to dispersing a solid into a liquid with or without stirring until a liquid is obtained, either a single phase, a layered heterogeneous solution, or a cloudy solution or suspension that is not completely transparent.

[0032] The term "volatile substances" refer to substances with low boiling points, especially organic solvents and water, such as acetonitrile, acetone, ethanol, isopropanol, ethyl acetate, hexane, cyclohexane, petroleum ether, methyl ethyl ketone, or their mixtures.

[0033] The term "salt" of a compound refers to a pharmaceutically acceptable salt with the desired pharmacological activity of the parent compound. Such salts include those formed as follows: when the acid protons in the parent compound are replaced by metal ions, such as alkali metal ions, alkaline earth metal ions, zinc or aluminum ions; or when the acid protons in the parent compound are coordinated with organic bases, such as glutamic acid, lysine, ethanolamine, diethanolamine, triethanolamine, trimethylaminomethane, *N*-methylglucamine and the like.

[0034] The compounds disclosed herein are generally named according to the IUPAC or CAS nomenclature system.

The crystal or crystal complex.

[0035] Due to the specificity of N-phosphoryl and O-acetyl phosphoric acid in its structure, contezolid acefosamil is unstable under some conditions as a prodrug, and conventional heating and dissolving, cooling and crystallization methods often lead to decomposition. A novel crystal or crystal complex of contezolid acefosamil (formula I) is disclosed herein. The crystal is stable at room temperature, and the nature of the crystal is obvious.

[0036] In formula I, R is sodium, ONa and O⁻Na⁺ are equivalent.

[0037] Disclosed is the crystal or crystal complex as a mixture of formula I with R=hydrogen and R=sodium each in any amount from 0 to 50%.

[0038] Disclosed is the content of the compound of formula I with R = hydrogen being less than 5% (by weight). This can improve the crystallinity of the compound.

[0039] Disclosed is the content of sodium in the crystal or crystal complex from 1 to 10% (by weight), more preferably from 3.2 to 6% (by weight) This can improve the crystallinity of the compound.

[0040] The crystal or crystal complex of the present disclosure exhibits more than one sharp diffraction peak between 0-40°2θ in the X-ray powder diffraction analysis (XRPD) (FIG. 2).

[0041] In some embodiments, the crystal or crystal complex exhibits at least one diffraction peak at about 5.0~40°2θ in the X-ray powder diffraction pattern.

[0042] In some embodiments, the crystal or crystal complex exhibits at least two diffraction peaks at about 20.0~25°2θ in the X-ray powder diffraction pattern.

[0043] In some embodiments, the crystal or crystal complex exhibits at least one sharp diffraction peak at about 6.0~8.0°2θ in the X-ray powder diffraction pattern.

[0044] In some embodiments, the crystal or crystal complex exhibits diffraction peaks at about 20.0~21.0°2θ in the X-ray powder diffraction pattern.

[0045] In some embodiments, the crystal or crystal complex exhibits diffraction peaks at about 23.0~24.0°2θ in the X-ray powder diffraction pattern.

[0046] The crystal or crystal complex of the present disclosure greatly improves the stability of contezolid acefosamil, for example, enabling it to be substantially free of decomposition at room temperature for 24 hours. The increase in the stability of the sample also proves the formation of the crystal or crystal complex.

[0047] In some embodiments, the crystal or crystal complex has a melting point of 220±10 °C.

Preparation

[0048] The preparation method of the above-mentioned crystal or crystal complex is also disclosed herein. In some

embodiments, the compound of formula I (contezolid acefosamil of formula I as a crude product) is dissolved in a biphasic solution of an organic solvent and water; inorganic salt is added, and then the product phase is separated to remove part or all of the volatile substances, and organic solvent is added, so as to obtain a crystal via crystallization. Contezolid acefosamil of formula I as a crude product can be prepared by the method disclosed in CN105612166A.

[0049] The volume ratio of organic solvent and water is preferably 1:9 to 4:6.

[0050] The organic solvent is selected from at least one of acetonitrile, acetone, methanol, ethanol, propanol, isopropanol, butanol, and tetrahydrofuran.

[0051] In a more preferred embodiment, the organic solvent may be selected from at least one of acetonitrile, ethanol, butanol, isopropanol, acetone, and tetrahydrofuran.

[0052] The method of dissolving the compound of formula I in the mixed solution of an organic solvent and water may be dissolving by stirring.

[0053] An inorganic salt is also added to the mixed solution of organic solvent and water. The inorganic salt is at least one of sodium chloride, sodium bromide, sodium iodide, sodium citrate, sodium dihydrogen phosphate, disodium hydrogen phosphate, sodium phosphate, sodium sulfate, sodium carbonate, sodium bicarbonate, sodium acetate and hydrates thereof.

[0054] During the crystallization process, the concentration of the crystallization solution, the rate of crystallization and temperature can be controlled.

[0055] Optionally, part or all of the volatile substances in the product phase such as organic solvent and water is removed. It can be removed by, for example, azeotropic distillation, vacuum distillation, and the like.

[0056] A second organic solvent is added to the product phase for crystallization. The second organic solvent is selected from at least one of acetonitrile, acetone, ethanol, and isopropanol. The quantity of the second organic solvent added can be 2-50 times the weight of the substrate.

[0057] Optionally, the product phase is cooled to about -30 °C to about 20 °C for crystallization.

[0058] After crystallization, the product can be separated out to obtain crystal or crystal complex. Further, the obtained crystal or crystal complex can be dried.

[0059] In some embodiments, the crystal is physically separated or filtered.

[0060] In some embodiments, the separated crystal is vacuum dried to obtain crystal or crystal complex. The temperature during vacuum drying can be about 15 °C to about 80 °C, preferably about 20 °C to about 50 °C.

[0061] In some embodiments, a method of treating a mammalian microbial infection or bacterial infection with the above-mentioned crystal or crystal complex is provided, comprising: administering a therapeutically effective amount of the crystal or crystal complex to the mammal. The crystal or crystal complex can be administered to mammals by oral, parenteral, transdermal, topical, rectal or intranasal routes in the form of a pharmaceutical composition. In some embodiments or in any embodiment, the microorganisms are Gram-positive microorganisms in the method. Therefore, the crystal or crystal complex of the present invention is a useful antimicrobial agent, and may be effective against many human and animal pathogens, including Gram-positive aerobic bacteria such as multi-resistant *Staphylococcus aureus*, *Enterococci*, and *Streptococci*, as well as anaerobic microorganisms such as *Bacteroides* and *Clostridium*, and acid-fast microorganisms such as *Mycobacterium tuberculosis* and *Mycobacterium avium*.

[0062] The crystal or crystal complex provided in the present disclosure can be conveniently added to the pharmaceutical composition in a conventional manner. The pharmaceutical composition or medicament may further comprise a pharmaceutically acceptable diluent, excipient, disintegrant, lubricant or carrier.

[0063] Pharmaceutical compositions containing the crystal or crystal complex of the present disclosure are prepared and the pharmaceutically acceptable carrier can be either solid or liquid. Solid form preparations include powders, tablets, dispersible granules, capsules, cachets, and suppositories. Tablets, powders, cachets, and capsules can be used as solid dosage forms suitable for oral administration. Illustrative examples of pharmaceutically acceptable formulations and methods for preparing certain pharmaceutical compositions are described, for example, in "The Science and Practice of Pharmacy," edited by A. Gennaro, Lippincott Williams & Wilkins, Baltimore, Maryland (Md., 2000).

[0064] In some embodiments, the crystal or crystal complex can be made into various pharmaceutically acceptable compositions such as water, a D5W injection solution or other tablets for treating or preventing various infections.

[0065] Hereinafter, the present invention will be further described with the following examples. It should be understood that the following examples are used to explain this invention but do not mean to limit the scope of this invention. Any non-essential improvements and modifications made by a person skilled in the art based on this invention all fall into the protection scope of this invention as defined by the claims. The specific process parameters below are only exemplary, and a person skilled in the art can choose proper values within an appropriate range according to the description, and are not restricted to the specific values shown below.

[0066] The test methods in the following embodiments are as follows:

X-ray powder diffraction analysis: Panalytical's XPERT-3 X-ray diffractometer was used. Approximately 10 mg samples were spread evenly on a sample tray of monocrystalline silicon XRPD tests were performed with the parameters described below:

Start Position [°2Th.]:	3.0121	End Position [°2Th.]:	39.9870
Step Size [°2Th.]:	0.0163	Scan Step Time [s]:	46.665
K-Alpha-1 [Å]:	1.54060	K-Alpha-2 [Å]:	1.54443
Generator Settings:	40mA, 45 kV		

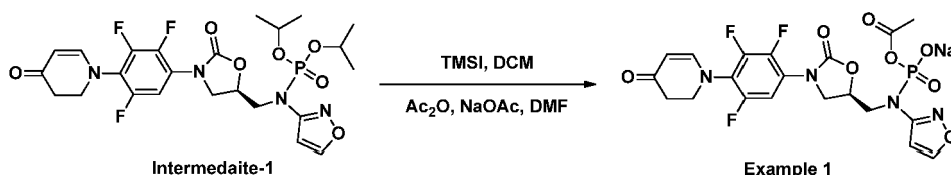
DSC analysis: TA Q2000 differential scanning calorimeter was used to perform DSC analysis with the following parameters:

Sample tray	Aluminum plate, gland
temperature range /°C	25-350 °C
Scan speed	10
Protective gas	Nitrogen

Example 1

Synthetic route:

[0067]



Compound of Example 1: Under nitrogen protection, trimethylsilyl iodide (14.4 g) was added dropwise to a solution of intermediate 1 (10.5 g, prepared according to the method of CN105612166A, Intermediate 2) in DCM (105 mL) at 0-10 °C, and stirred at room temperature for 2 hours. The reaction solution was concentrated under reduced pressure, and the residue was rinsed with methyl tert-butyl ether (100 mL), filtered, and dried in vacuo. The resulting product was redissolved in DMSO/MeCN (10.5 mL/105 mL), to which NaOAc (36.7 g) and Ac₂O (5.9 g) were added, and the mixture was stirred for 1 hour. MTBE (700 mL) was added, and the mixture was stirred, filtered, and the filter residue was dried in vacuo to obtain crude compound of Example 1.

[0068] Example 1 Product Form 1: The crude product of Example 1 was stirred in 5% EtOH/DCM (1000 mL), and filtered, and the mother liquor was concentrated under reduced pressure. The residue was slurried with ethyl acetate:methyl tert-butyl ether (3:1, 300 mL). Filtration, washing (the residue was washed with ethyl acetate: methyl tert-butyl ether), and vacuum drying gave a yellow solid as product form 1. MS: 531 [M + H]. The product X-ray powder diffraction analysis of the product was shown in FIG. 3.

[0069] Example 1 Product Form 2: The crude product of Example 1 was dissolved in water and purified by reverse phase HPLC (C18) gradient. The mobile phase was water and acetonitrile. The fractions containing Example 1 were collected. Lyophilization gave a white solid as product form 2. MS: 531 [M+H]. The product X-ray powder diffraction analysis of the product was shown in FIG. 1.

[0070] Crystal or crystal complex of the product of Example 1 (product form 3): The crude product of Example 1 (146 g) was dissolved in water (582 mL) and acetonitrile (169 mL) at room temperature, and dissolved by stirring, and sodium dihydrogen phosphate dihydrate (349 g) was added and dissolved by stirring. The product phase was separated and concentrated under reduced pressure until there was no obvious fraction, and then acetone (550 mL) was added to dissolve the residue. The resulting mixture was cooled, and stirred for crystallization followed by filtration and vacuum drying to obtain a white solid with a sodium content of 4.3%. The product X-ray powder diffraction analysis of the product was shown in FIG. 2, and the DSC analysis result was shown in FIG. 4.

Test and Application

[0071] In the early development and research of contezolid acefosamil, the product obtained via laboratory's traditional operations such as extraction, and evaporation under reduced pressure were unstable in storage at room temperature. Even the product lyophilized by HPLC purification was not stable enough, while X-ray powder diffraction showed no diffraction peaks. Surprisingly, for the product obtained by the preparation method of the present invention, the X-ray powder diffraction analysis showed many sharp diffraction peaks between $0-40^{\circ}2\theta$, and the analysis result of the differential scanning calorimetry (DSC) also showed a narrow melting range, indicating the crystal features. More importantly, the resulting crystalline product had better stability than amorphous powder. The detailed data are as follows.

[0072] The X-ray powder diffraction data of the product obtained by the preparation method of the present invention are shown in Table 1 and the diffraction analysis chart is shown in FIG. 2.

Table 1 shows main peaks of crystal X-ray powder diffraction of the present invention.

angle ($2\theta^{\circ}$)	relative strength (%)
6.9	100
14.7	29.5
15.5	29.3
16.5	31.2
20.2	50.8
22.9	42.1
23.7	54.4

[0073] It can be seen from Table 1 and FIG. 2 that the product has more than one distinct sharp peak. This shows that the product has obvious crystallographic features that reflect X-rays on the crystal surface. It is a new crystal that cannot be obtained by routine methods of manipulation by solvent recrystallization, HPLC purification, or lyophilization.

[0074] The stability of the crystal of the present invention: The purity change (stability) of the two samples at room temperature for nine months was determined by the HPLC method. One was the product obtained by HPLC purification lyophilization (product form 2 in Example 1); the other was the product (product form 3 in Example 1) obtained by the method which was provided by the present invention for preparing the crystal. The results are shown in Table 2.

Table 2 shows stability of the product for nine months (HPLC purity, %).

	Time (month)				
	0	1	2	3	9
Product form 2	97.29	94.27	92.99	91.27	83.21
Product form 3	97.54	96.61	96.77	96.84	95.81

[0075] It can be seen from Table 2 that both products decompose slowly. Product form 3 (crystal) is significantly more stable than product form 2 (amorphous powder) at room temperature, and more pronounced at nine months (95.81 % and 83.21 %). The large increase in the stability of the sample also confirms that the existence of the crystal or crystal complex.

[0076] On the other hand, the analysis results of differential scanning calorimetry (DSC) also confirms the existence of crystalline material. FIG. 5 is a DSC analysis diagram of the crystal (product form 3 in Example 1) that was re-dissolved in water and lyophilized.

[0077] In comparison with the DSC analysis diagram of the crystal in FIG. 4, the product obviously has a narrow melting range indicating a crystal feature of a single crystal or of a crystallized complex with a crystal dominant component.

[0078] The method for preparing the crystal form of the present invention is unique and cannot be achieved by traditional crystallizing operations (heating solvent and dissolution, cooling and crystallization). For example, acetonitrile, one of the solvents used for extraction in some embodiments, is not used in general extraction operations.

[0079] The crystal or crystal complex of the oxazolidinone compounds of the present invention shows effective in vivo activity against a variety of microorganisms, including Gram-positive microorganisms. In order to determine the effective therapeutic activity of the crystal or crystal complex of the present invention, a mouse peritonitis infection model according

to the general procedure described by Marra et al., Current Protocols in Pharmacology (2005), 13A.4.1-13A.4.13 was used as the test method.

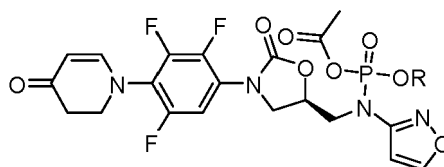
[0080] In this infection model, pathogenic *Staphylococcus aureus* strain SAU1018 was used, and an intravenous injection of a saline solution containing these crystals suitable for clinical or therapeutic use was administered to infected animals, showing high in vivo activity. The ED₅₀ value (effective dose for 50% animal survival in the test) of the crystal or crystal complex is 10 mg/kg.

[0081] The crystal or crystal complex of the present invention can also be administered using a convenient oral route. When administered orally to the above-mentioned mouse model animals infected with *Staphylococcus aureus*, the compound showed high antibacterial efficacy, and its ED₅₀ value was about 8 mg/kg.

[0082] Accordingly, the crystals of the present invention or substance that contains the crystal and composition thereof of the present invention possess useful antibacterial activity. Therefore, the crystals of the present invention are useful antimicrobial agents and may be effective against many human and animal pathogens, including Gram-positive aerobic bacteria such as multi-resistant *Staphylococcus aureus*, Enterococci, Streptococci, and anaerobic microorganisms such as *Bacteroides* and *Clostridium*, and acid-fast microorganisms such as *Mycobacterium tuberculosis* and *Mycobacterium avium*. These pharmaceutical compositions can be administered by a variety of different routes such as intravenous injection, oral administration, parenteral administration, transdermal administration, topical administration, rectal administration and intranasal administration.

Claims

1. A crystal or crystal complex of formula I, wherein R is sodium,



I

wherein the crystal or crystal complex exhibits diffraction peaks at $6.9^{\circ} \pm 0.2^{\circ}$, $14.7^{\circ} \pm 0.2^{\circ}$, $15.5^{\circ} \pm 0.2^{\circ}$, $16.5^{\circ} \pm 0.2^{\circ}$, $20.2^{\circ} \pm 0.2^{\circ}$, $22.9^{\circ} \pm 0.2^{\circ}$, and $23.7^{\circ} \pm 0.2^{\circ}$ 2θ in its X-ray powder diffraction (XRPD) pattern.

2. The crystal or crystal complex of claim 1, wherein the crystal or crystal complex has a melting point of $220 \pm 10^{\circ}\text{C}$.

3. A method for preparing the crystal or crystal complex according to claim 1 or 2, comprising the following steps:

- (1) preparing a compound of formula I as a crude product;
- (2) dissolving the compound of formula I as a crude product in a mixed solution of an organic solvent and water;
- (3) adding inorganic salt to the mixture, separating the product phase to remove part or all of the volatile substances, and adding a second organic solvent, so as to obtain a crystal via crystallization; and
- (4) separating out the crystals of the compound of formula I, wherein in step (2), the organic solvent is selected from at least one of acetonitrile, acetone, methanol, ethanol, propanol, isopropanol, butanol, and tetrahydrofuran, wherein in step (3), the inorganic salt is selected from at least one of sodium chloride, sodium bromide, sodium iodide, sodium citrate, sodium dihydrogen phosphate, disodium hydrogen phosphate, sodium phosphate, sodium sulfate, sodium carbonate, sodium bicarbonate, sodium acetate and their hydrates, and wherein in step (3), the second organic solvent is selected from at least one of acetonitrile, acetone, ethanol, and isopropanol.

4. The method of claim 3, wherein, in step (2) the volume ratio of the organic solvent and water is 1:9 to 4:6.

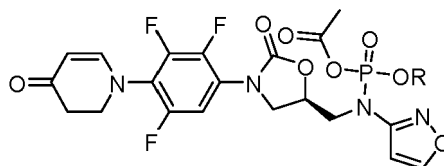
5. A pharmaceutical composition comprising the crystal or crystal complex according to claim 1 or 2 and a pharmaceutically acceptable carrier.

6. A crystal or crystal complex according to claim 1 or 2 or a pharmaceutical composition of claim 5 for use in the treatment of a microbial infection in a mammal.

7. The crystal, crystal complex or pharmaceutical composition for use according to claim 6, wherein the crystal, crystal complex or pharmaceutical composition is for use in treating a bacterial infection in a human.
8. The crystal, crystal complex or pharmaceutical composition for use according to claim 6 or 7, wherein the bacterial infection is a Gram-positive bacterial infection.

Patentansprüche

1. Kristall oder Kristallkomplex der Formel I, wobei R Natrium ist,



I

wobei der Kristall oder Kristallkomplex Beugungspicks bei $6,9^\circ \pm 0,2^\circ$, $14,7^\circ \pm 0,2^\circ$, $15,5^\circ \pm 0,2^\circ$, $16,5^\circ \pm 0,2^\circ$, $20,2^\circ \pm 0,2^\circ$, $22,9^\circ \pm 0,2^\circ$ und $23,7^\circ \pm 0,2^\circ$ 2θ in seinem Röntgenpulverbeugungs(XRPD)-Muster aufweist.

2. Kristall oder Kristallkomplex nach Anspruch 1, wobei der Kristall oder Kristallkomplex einen Schmelzpunkt von $220 \pm 10^\circ\text{C}$ aufweist.
3. Verfahren zur Herstellung des Kristalls oder Kristallkomplexes nach Anspruch 1 oder 2, umfassend die folgenden Schritte:

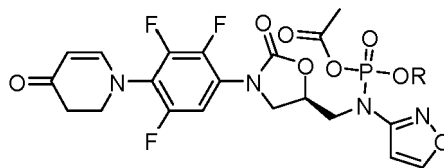
- (1) Herstellen einer Verbindung der Formel I als Rohprodukt;
- (2) Auflösen der Verbindung der Formel I als Rohprodukt in einer gemischten Lösung aus einem organischen Lösungsmittel und Wasser;
- (3) Zugabe von anorganischem Salz zu dem Gemisch, Trennen der Produktphase, um einen Teil oder alle der flüchtigen Substanzen zu entfernen, und Zugabe eines zweiten organischen Lösungsmittels, um so einen Kristall durch Kristallisation zu erhalten; und
- (4) Heraustrennen der Kristalle der Verbindung der Formel I, wobei, in Schritt (2), das organische Lösungsmittel ausgewählt ist aus mindestens einem von Acetonitril, Aceton, Methanol, Ethanol, Propanol, Isopropanol, Butanol und Tetrahydrofuran, wobei, in Schritt (3), das anorganische Salz ausgewählt ist aus mindestens einem von Natriumchlorid, Natriumbromid, Natriumiodid, Natriumcitrat, Natriumdihydrogenphosphat, Dinatriumhydrogenphosphat, Natriumphosphat, Natriumsulfat, Natriumcarbonat, Natriumbicarbonat, Natriumacetat und deren Hydraten, und wobei, in Schritt (3), das zweite organische Lösungsmittel ausgewählt ist aus mindestens einem von Acetonitril, Aceton, Ethanol und Isopropanol.

4. Verfahren nach Anspruch 3, wobei, in Schritt (2), das Volumenverhältnis des organischen Lösungsmittels und Wassers 1:9 bis 4:6 beträgt.
5. Pharmazeutische Zusammensetzung, umfassend den Kristall oder Kristallkomplex gemäß Anspruch 1 oder 2 und einen pharmazeutisch verträglichen Träger.

6. Kristall oder Kristallkomplex nach Anspruch 1 oder 2 oder pharmazeutische Zusammensetzung nach Anspruch 5 zur Verwendung bei der Behandlung einer mikrobiellen Infektion bei einem Säugetier.
7. Kristall, Kristallkomplex oder pharmazeutische Zusammensetzung zur Verwendung nach Anspruch 6, wobei der Kristall, der Kristallkomplex oder die pharmazeutische Zusammensetzung zur Verwendung bei der Behandlung einer bakteriellen Infektion bei einem Menschen dient.
8. Kristall, Kristallkomplex oder pharmazeutische Zusammensetzung zur Verwendung nach Anspruch 6 oder 7, wobei die bakterielle Infektion eine grampositive bakterielle Infektion ist.

Revendications

1. Cristal ou complexe cristallin de formule I, dans laquelle R représente un atome de sodium,



I

dans laquelle le cristal ou le complexe cristallin présente des pics de diffraction à $6,9^\circ \pm 0,2^\circ$, $14,7^\circ \pm 0,2^\circ$, $15,5^\circ \pm 0,2^\circ$, $16,5^\circ \pm 0,2^\circ$, $20,2^\circ \pm 0,2^\circ$, $22,9^\circ \pm 0,2^\circ$, et $23,7^\circ \pm 0,2^\circ$ 2θ dans son diagramme de diffraction des rayons X sur poudre (XRPD).

2. Cristal ou complexe cristallin selon la revendication 1, ledit cristal ou complexe cristallin comportant un point de fusion de $220 \pm 10^\circ\text{C}$.

3. Procédé de préparation du cristal ou complexe cristallin selon la revendication 1 ou 2, comprenant les étapes suivantes :

- (1) la préparation d'un composé de formule I sous forme de produit brut ;
- (2) la dissolution du composé de formule I sous forme de produit brut dans une solution mixte d'un solvant organique et d'eau ;
- (3) l'ajout d'un sel inorganique au mélange, la séparation de la phase du produit pour enlever une partie ou la totalité des substances volatiles, et l'ajout d'un second solvant organique, de façon à obtenir un cristal par cristallisation ; et
- (4) la séparation des cristaux du composé de formule I, dans ladite étape (2) le solvant organique étant choisi parmi au moins un solvant parmi l'acétonitrile, l'acétone, le méthanol, l'éthanol, le propanol, l'isopropanol, le butanol et le tétrahydrofurane, dans ladite étape (3) le sel inorganique étant choisi parmi au moins un sel parmi le chlorure de sodium, le bromure de sodium, l'iodure de sodium, le citrate de sodium, le dihydrogénophosphate de sodium, l'hydrogénophosphate de disodium, le phosphate de sodium, le sulfate de sodium, le carbonate de sodium, le bicarbonate de sodium, l'acétate de sodium et leurs hydrates, et dans ladite étape (3) le second solvant organique étant choisi parmi au moins un solvant parmi l'acétonitrile, l'acétone, l'éthanol et l'isopropanol.

4. Procédé selon la revendication 3, dans ladite étape (2) le rapport volumique du solvant organique et de l'eau étant de 1:9 à 4:6.

5. Composition pharmaceutique comprenant le cristal ou le complexe cristallin selon la revendication 1 ou 2 et un vecteur pharmaceutiquement acceptable.

6. Cristal ou complexe cristallin selon la revendication 1 ou 2 ou composition pharmaceutique selon la revendication 5 destiné à être utilisé dans le traitement d'une infection microbienne chez un mammifère.

7. Cristal, complexe cristallin ou composition pharmaceutique destiné à être utilisé selon la revendication 6, ledit cristal, ledit complexe cristallin ou ladite composition pharmaceutique étant destiné à être utilisé dans le traitement d'une infection bactérienne chez un humain.

8. Cristal, complexe cristallin ou composition pharmaceutique destiné à être utilisé selon la revendication 6 ou 7, ladite infection bactérienne étant une infection bactérienne à Gram positif.

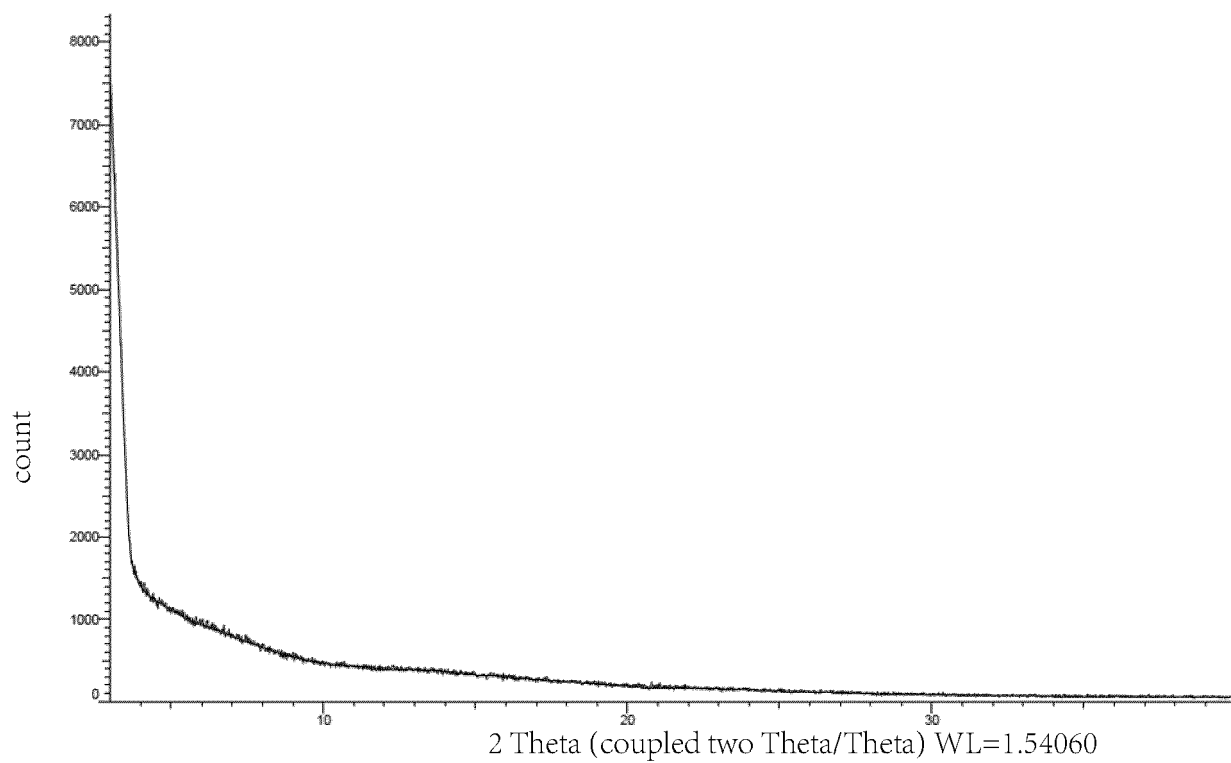


FIG. 1

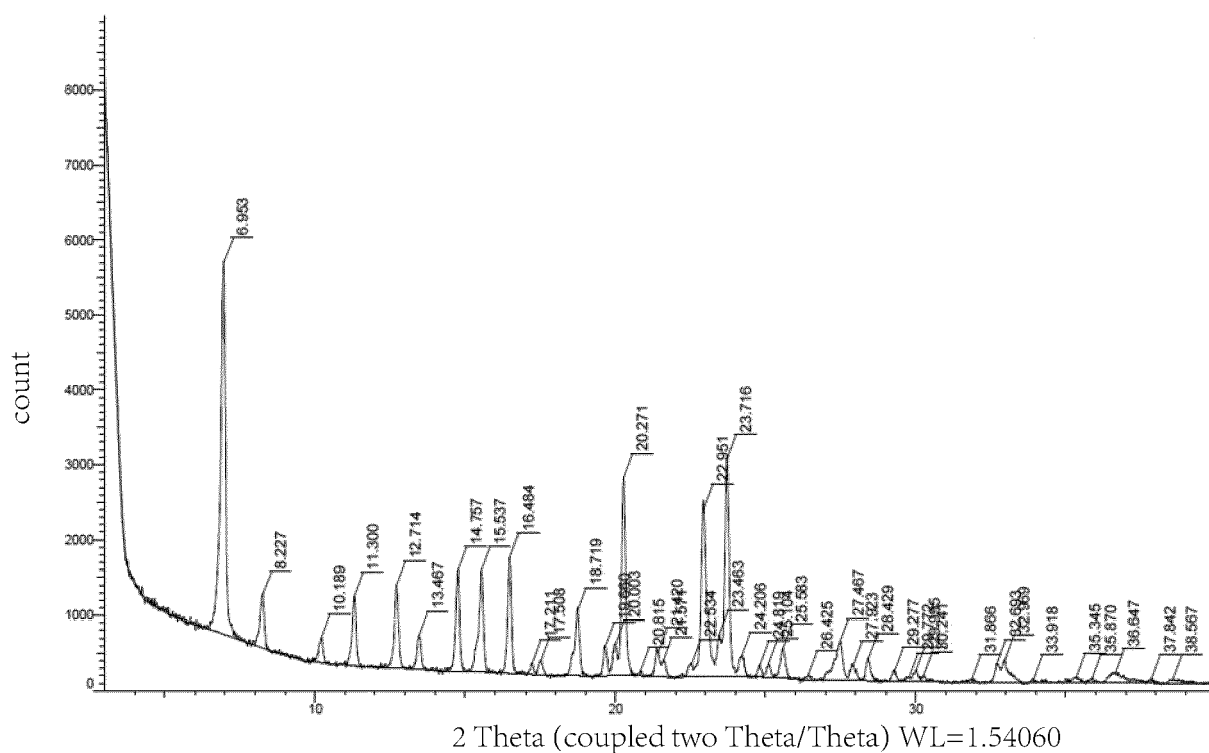


FIG. 2

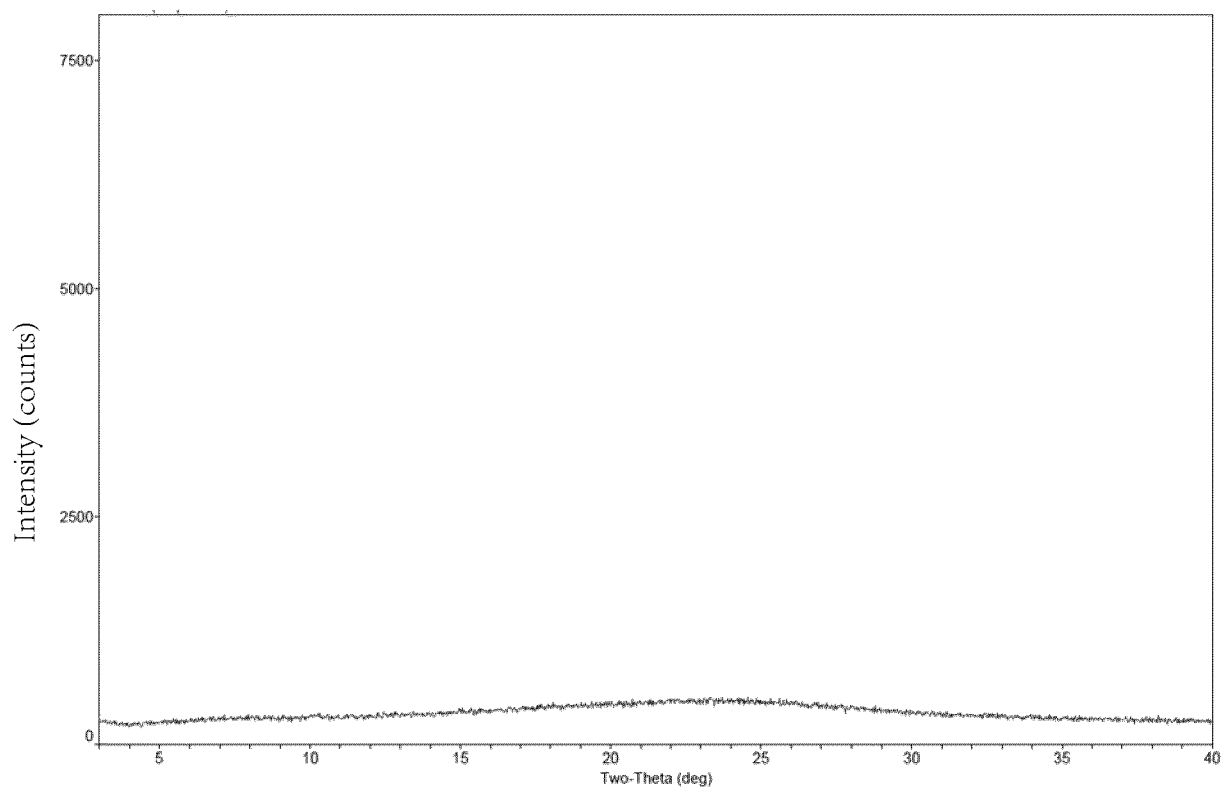


FIG. 3

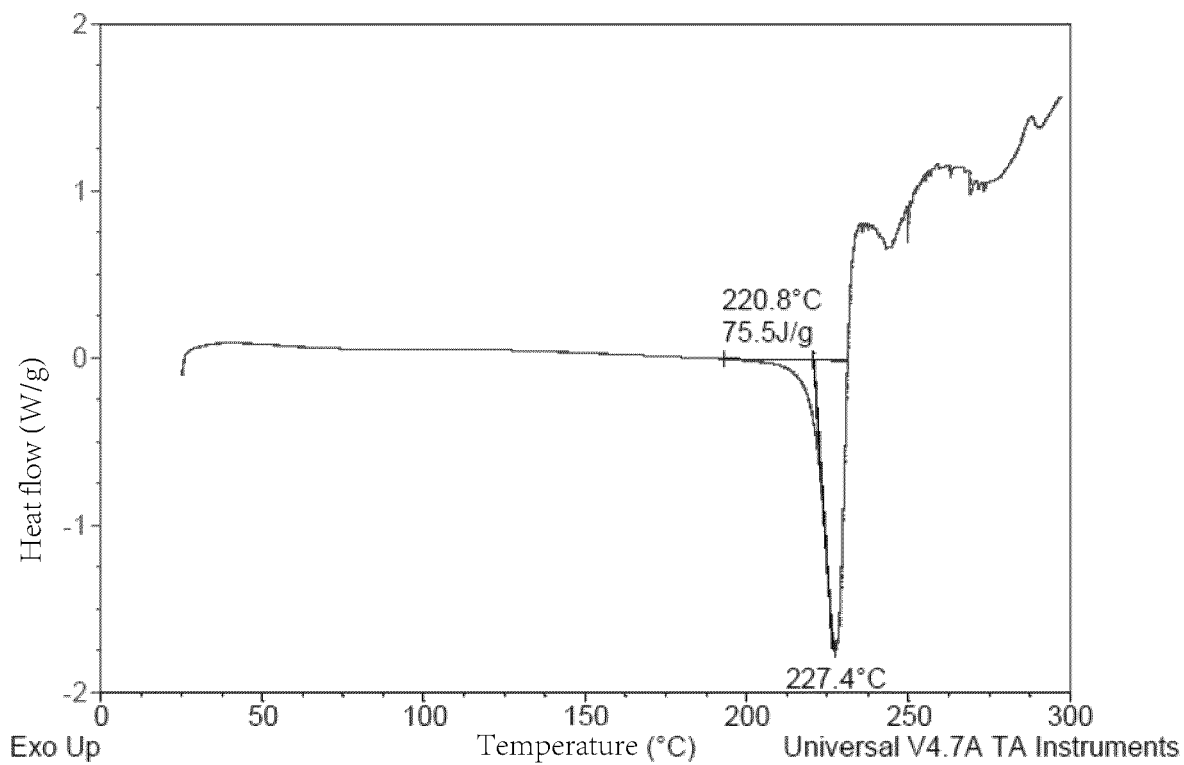


FIG. 4

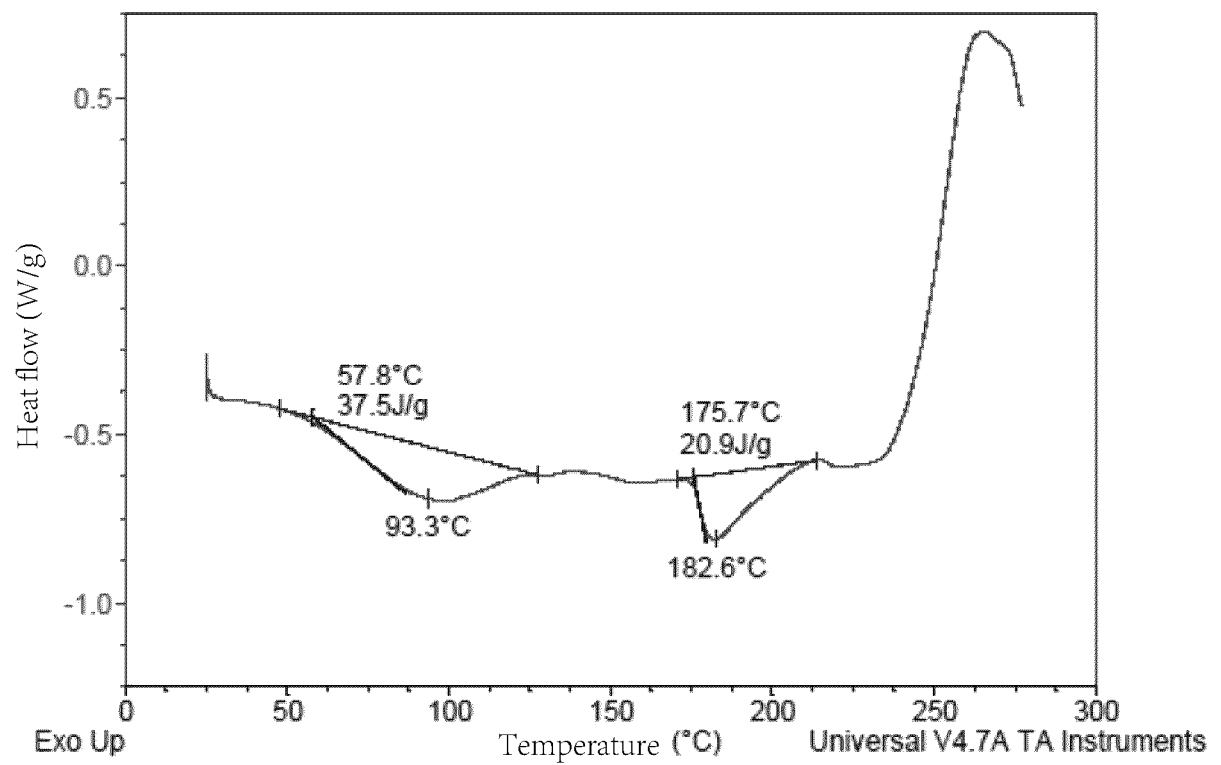


FIG. 5

REFERENCES CITED IN THE DESCRIPTION

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